Experiment 1

Numerical Study of Flow Around a NACA 0012 Airfoil

Aim

To examine how the angle of attack affects the flow field, wake, convergence behaviour, and qualitative lift and drag around a NACA 0012 airfoil.

Simulation Parameters

Parameter	Value
Streamwise grid points	130
Cross-stream grid points	70
Reynolds number	120
Angle of attack	9°
Time step	3.623×10^{-3}
Iterations	300
Poisson iterations per step	80
Chord length / leading-edge position	1.0 / x = 1.35

Numerical Method

The incompressible Navier–Stokes equations are solved with a Cartesian immersed-geometry projection approach. The airfoil is represented by a smooth solid mask; velocity penalisation is applied within the body and a pressure-Poisson correction enforces near-zero divergence. The time step is chosen from convective and viscous stability limits.

Boundary Conditions

Uniform inflow and far-field conditions are applied at the inlet and outer boundaries. The outlet uses a convective or zero-gradient treatment with a pressure reference. A no-slip condition is imposed on the airfoil through the immersed mask.

Observations

Continuity residual, maximum speed, wake vorticity and qualitative lift/drag histories were examined for the 9° case. The continuity residual fell from 3.89×10^{-1} at iteration 1 to 7.71×10^{-2} at iteration 60, then levelled at about 6.3×10^{-2} (6.285×10^{-2} at iteration 300). The maximum speed rose from 1.083 to 1.242 over the run.

Results

Angle of attack	Drag coefficient (qualitative)	Lift coefficient (qualitative)	Final continuity L_2

9°	0.40	−0.60	6.29×10^{-2}

Iteration history of the force coefficients:

Iteration	**Continuity L_2**	**Max speed**	**Cd**	**Cl**
1	3.890e-01	1.083	2.771	−4.149
60	7.705e-02	1.219	0.628	−0.772
120	6.452e-02	1.231	0.488	−0.604
180	6.264e-02	1.237	0.437	−0.587
240	6.260e-02	1.240	0.414	−0.594
300	6.285e-02	1.242	0.403	−0.600

Conclusion

The continuity residual drops quickly during the early iterations and then flattens, indicating that the solution approaches a quasi-steady numerical state, although the plateau ($\sim 6 \times 10^{-2}$) shows the divergence is reduced but not driven to a small value. The vorticity plot shows a shear layer and a concentrated wake behind the airfoil. The force coefficients settle at Cd ≈ 0.40 and Cl ≈ −0.60.

These coefficients come from an immersed-boundary penalty-force approximation, so they are qualitative indicators, not validated aerodynamic predictions. The negative lift at a positive 9° incidence is physically unexpected and should be checked against the force sign convention and the flow-field orientation before any physical conclusion is drawn.

Original Simulation Figures

The following figures are the plots and simulation outputs generated by the notebook for the 9° case.

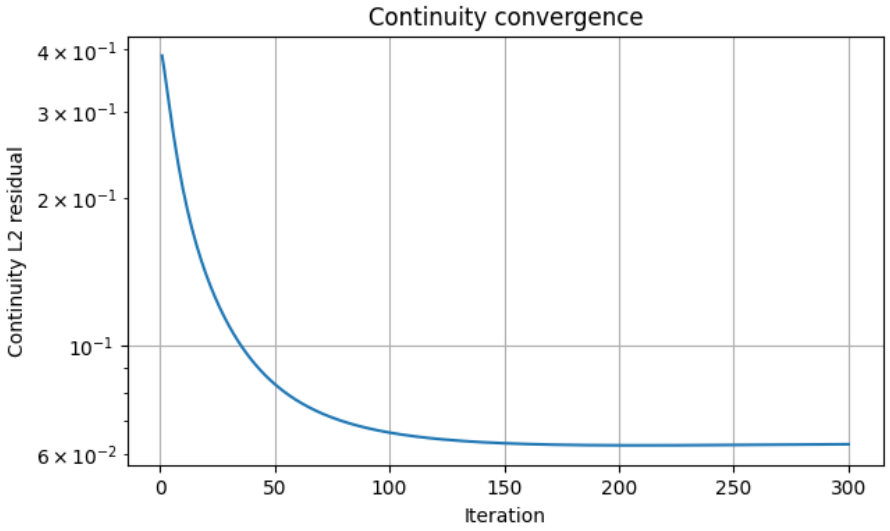


Figure 1 Continuity convergence with iterations (α = 9°)

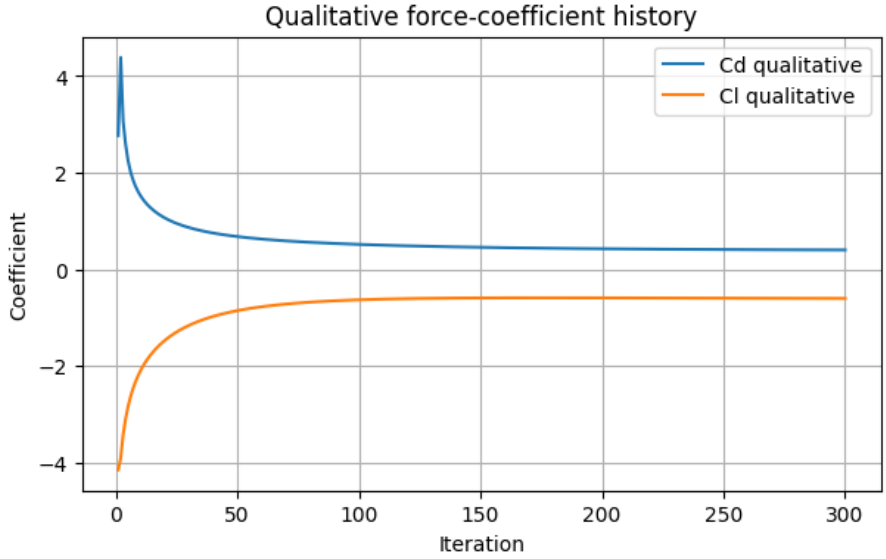


Figure 2 Qualitative force-coefficient history (Cd and Cl) with iterations (α = 9°)

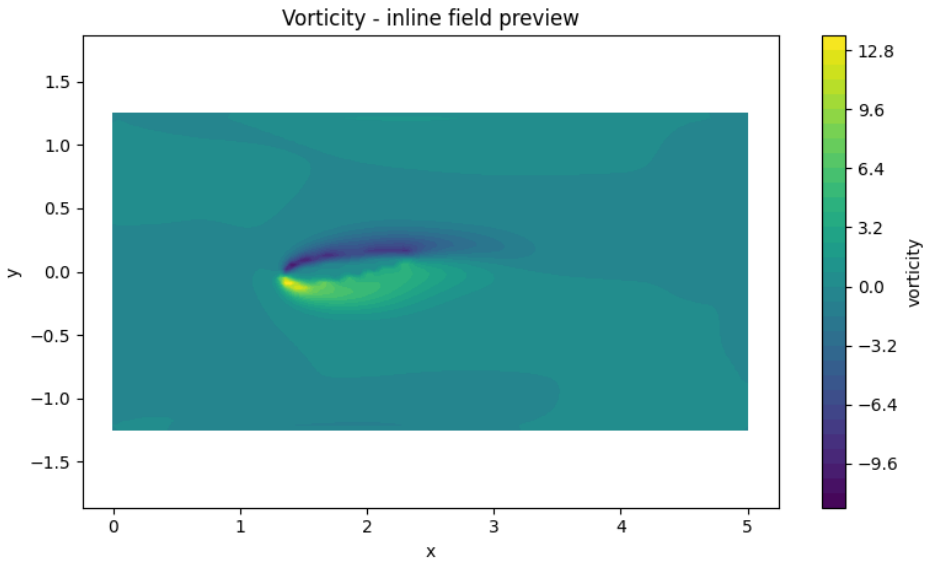


Figure 3 Vorticity field: wake visualization (α = 9°)